Module-1

Principles of Wireless Communications: The Wireless Communication Environment, Modelling of wireless systems, System model for narrowband Signals, Rayleigh fading Wireless Channel.

The Wireless Channel: Basics of Wireless Channel Modelling, Average Delay Spread in Outdoor Cellular Channels, Coherence bandwidth, Relation between ISI and Coherence Bandwidth, Doppler fading, Doppler Impact on a wireless Channel, Coherence Time.

[Text1: 3.1 to 3.4, 4.1 to 4.7]

Questions

1. Define multipath propagation. Explain the wireless system model for base band and narrow band signal. (10 marks)
2. Define wireless communication system. Explain Rayleigh Fading wireless channels. (10 marks)
3. Explain the basics of wireless channel modelling (10 marks)
   Or
   Explain a) Delay spread
   b) Coherence bandwidth
   c) Doppler spread
   d) coherence Time
4. Explain the doppler impact and doppler fading in wireless channels. (10 marks)
5. Write note on coherence Time (10 marks)

1. Define multipath propagation. Explain the wireless system model for base band and narrow band signal.

<u>Principles of Wireless Communications</u>

In wireless systems, a signal can travel to the receiver via multiple paths (direct, reflected, scattered). This results in multiple copies of the signal, each with different delays, attenuations, and phase shifts, arriving at the receiver and causing **multipath interference**.

This interference leads to **multipath fading** (or simply **fading**), where the combined signal power can be amplified or attenuated. A **deep fade** is a strong destructive interference that causes a severe drop in signal power, potentially leading to communication failure.

<u>The system model for Baseband band signals</u>

The transmitted signal is represented as a passband signal

$$s(t) = \mathrm{Re}\left\{s_b(t)\, e^{j2\pi f_c t}\right\}$$

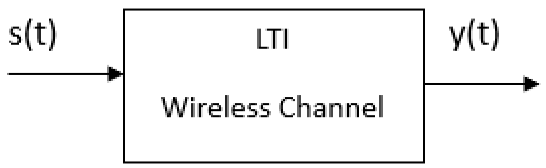
where sb(t) is the complex baseband signal and fc is the carrier frequency.

The wireless channel itself is modelled as a Linear Time-Invariant (LTI) system with L multipath components. Each path i is characterized by:

$$h_i(\tau) = a_i \delta(\tau - \tau_i).$$

Where ai : attenuation factor

τi: A path delay



The sum of all individual path responses:

$$h(\tau) = \sum_{i=0}^{L-1} a_i \delta(\tau - \tau_i)$$

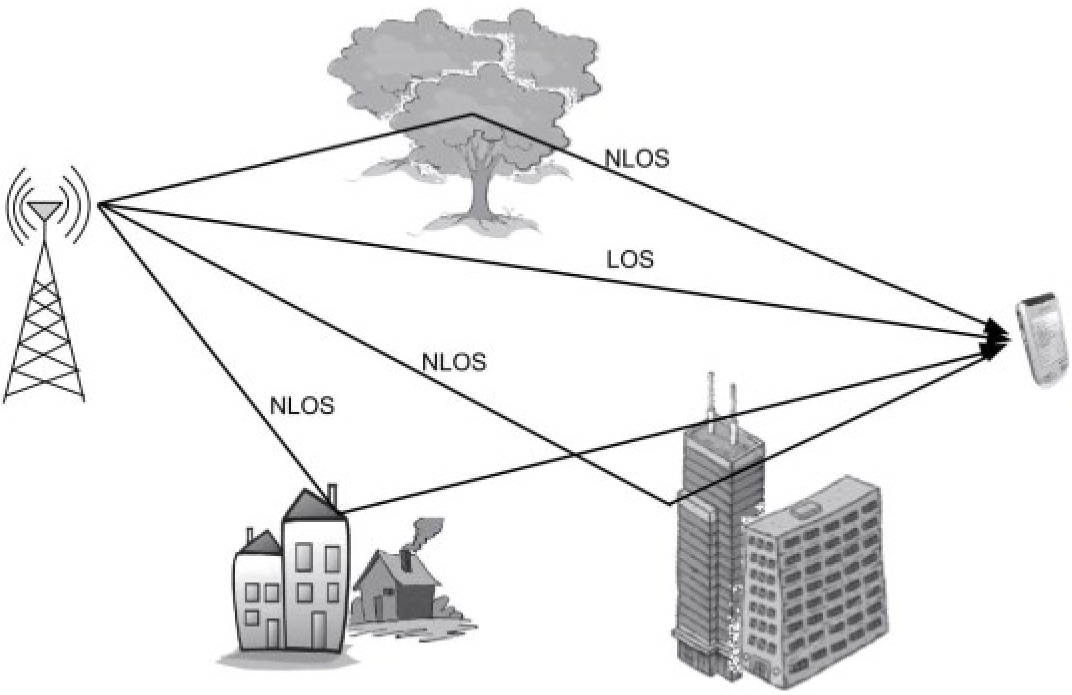


Figure 3.1 *Schematic of the wireless-propagation environment*

The paths can include (i) Line-Of-Sight (LOS)

(ii) Non-Line-Of-Sight (NLOS) (and reflected/scattered)

The received signal y(t) is derived by convolving the transmitted signal s(t)s(t) with the channel's impulse response h(τ).

$$y(t) = \sum_{i=0}^{L-1} a_i s(t - \tau_i)$$

Expressing this in terms of the complex baseband signal $s_b(t)$ yields the equivalent received baseband signal $y_b(t)$:

$$y_b(t) = \sum_{i=0}^{L-1} a_i e^{-j2\pi f_c \tau_i} s_b(t - \tau_i)$$

Where $e^{-j2\pi f c \tau i}$ is phase shift

An important channel parameter introduced is the **delay spread** Tm, which is the time difference between the first and last arriving signal copies. The text notes that for a **narrowband channel**, the delay spread is much smaller than the symbol time T (Tm<<T), which allows for further simplification of the model.

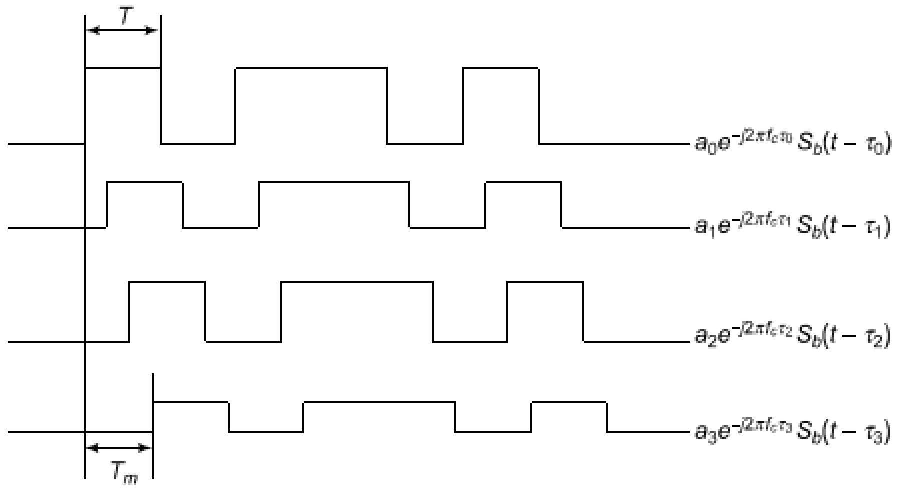


Figure 3.2 *Multipath signal components at the receiver*

<u>The system model for Narrow band signals</u>

$$y_b(t) = \underbrace{\left(\sum_{i=0}^{L-1} a_i e^{-j2\pi f_c \tau_i}\right)}_{ae^{j\phi}} s_b(t) = h s_b(t)$$

where $h = ae^{j\phi} = \sum_{i=0}^{L-1} a_i e^{-j2\pi f_c \tau_i}$ is termed the *complex* fading *channel coefficient*. Hence, the output baseband signal $y_b(t)$ is related to the input baseband signal $s_b(t)$ by a complex attenuation factor $ae^{j\phi}$.

Problems:

EXAMPLE 3.1

Consider a wireless signal with a carrier frequency of $f_c = 850$ MHz, which is transmitted over a wireless channel that results in $L = 4$ multipath components at delays of $201, 513, 819, 1223$ ns and corresponding to received signal amplitudes of $1, 0.6, 0.3, 0.2$ respectively. Derive the expression for the received baseband signal $y_b(t)$ if the transmitted baseband signal is $s_b(t)$.

Solution: As given in the expression for the received baseband signal in Eq. (3.3), we need to compute the factors $a_i e^{-j2\pi f_c \tau_i}$ for $i = 0, 1, 2, 3$ to derive the expression for the received baseband signal $y_b(t)$. Accordingly, the factor $a_0 e^{-j2\pi f_c \tau_0}$ is given as

$$a_0 e^{-j2\pi f_c \tau_0} = 1 \times e^{-j2\pi 850 \times 10^6 \times 201 \times 10^{-9}}$$

$$= 0.59 + j0.81$$

Similarly, the other factors for $i = 1, 2, 3$ can be computed as $0.57 - j0.19$, $0.18 - j0.24$, $-0.19 + j0.06$ respectively. Hence, the receive baseband signal is, therefore, given as

$$y_b(t) = (0.59 + j0.81)\, s_b(t) + (0.57 - j0.19)\, s_b\left(t - 201 \times 10^{-9}\right)$$

$$+ (0.18 - j0.24)\, s_b\left(t - 513 \times 10^{-9}\right) + (-0.19 + j0.06)\, s_b\left(t - 1223 \times 10^{-9}\right)$$

Thus, the receiver sees $L = 4$ signal copies $s_b(t - \tau_i)$, each delayed by τ_i, attenuated and phase shifted by $a_i e^{-j2\pi f_c \tau_i}$.

EXAMPLE 3.2

For the wireless channel given in Example 3.1, derive the corresponding received signal with the narrowband assumption.

Solution: As given in Eq. (3.4), the received signal with the narrowband assumption is given as

$$y(t) = \left(\sum_{i=0}^{3} a_i e^{-j2\pi f_c \tau_i}\right) s_b(t)$$

$$= (1.14 + j0.44)\, s_b(t)$$

Thus, the net received baseband signal in this case can be expressed as $y_b(t) = h s_b(t)$, where h the channel coefficient is $h = 1.14 + j0.44$.

2. Define wireless communication system. Explain Rayleigh Fading wireless channels. (10 marks)

A wireless communication system **is a system that transfers information (like voice, data, video) between two or more points** without the use of any physical connecting wires, cables, or waveguides.

<u>Rayleigh Fading wireless channel</u>

The complex fading coefficient h is represented as:

$$h = ae^{j\phi} = \sum_{i=0}^{L-1} (x_i + jy_i) = X + jY$$

where X and Y are the real and imaginary parts, resulting from the sum of many multipath components xi and yi.

- The **marginal PDF of the amplitude** A is obtained by integrating the joint PDF $f_{A,\Phi}(a,\phi)$ over all possible phases ϕ:

$$f_A(a) = \int_{-\pi}^{\pi} f_{A,\Phi}(a,\phi)\, d\phi = 2ae^{-a^2}, \quad 0 \leq a < \infty$$

This is the **Rayleigh distribution.**

- The **marginal PDF of the phase** Φ is found by integrating the joint PDF over all possible amplitudes a:

$$f_\Phi(\phi) = \int_{0}^{\infty} f_{A,\Phi}(a,\phi)\, da = \frac{1}{2\pi}, \quad -\pi < \phi \leq \pi$$

This is the **uniform distribution** over $[-\pi, \pi]$.

- The model is named **Rayleigh fading** because the amplitude A follows a Rayleigh distribution.
- The text also mentions that the next step is to find the **average power** in the amplitude a of the fading coefficient h, which is a key parameter in wireless communications.

- The model is now fully derived and is termed a **Rayleigh fading wireless channel**, named after the **Rayleigh distribution** of the amplitude A.
- The **average power** of the channel coefficient h is calculated as:

$$E\{|h|^2\} = E\{a^2\} = E\{X^2 + Y^2\} = 1$$

This shows that the average power is normalized to 1.

- Although the term "Rayleigh" specifically refers to the amplitude distribution, the **Rayleigh fading channel model** characterizes both:
  - The **amplitude** (Rayleigh-distributed)
  - The **phase** (uniformly distributed)

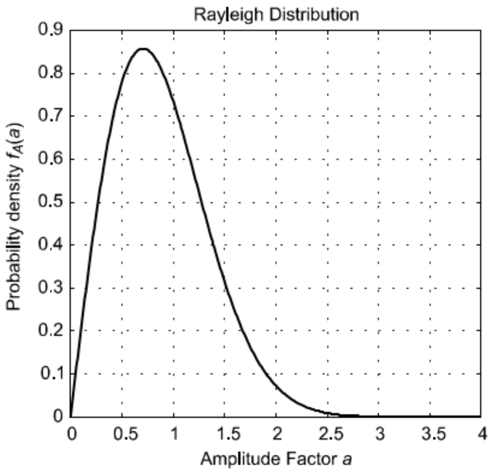


Figure 3.3 *Rayleigh density for amplitude factor a*

EXAMPLE 3.4

In the wireless Rayleigh fading channel described above, consider a transmit power $P_t\,(\mathrm{dB}) = 20$ dB. What is the probability that the power at the receiver is greater than $P_r\,(\mathrm{dB}) = 10$ dB ?

Solution: First, let us begin by computing the appropriate linear power values for the above given dB values. $P_t\,(\mathrm{dB}) = 10\log_{10}(P_t)$. Hence, the linear transmit power P_t is given as $P_t = 10^{P_t(\mathrm{dB})/10} = 10^2 = 100$. Similarly, the linear receiver power corresponding to $P_r\,(\mathrm{dB}) = 10$ dB is given as $P_r = 10^1 = 10$. Also, observe that given a power gain g, the received power is simply $P_r = gP_t$. Hence, for a received power $P_r > 10$, it naturally implies that

$$
\begin{aligned}
gP_t &> 10 \\
\Rightarrow g &> \frac{10}{P_t} = \frac{10}{20} \\
&= \frac{1}{2}
\end{aligned}
$$

Thus, the probability that the received power is greater than 10 essentially corresponds to the probability that the random power gain g of the Rayleigh fading wireless channel is greater than $\frac{1}{2}$. This probability can be readily computed as

$$
\begin{aligned}
\Pr\left(g > \frac{1}{2}\right) &= \int_{\frac{1}{2}}^{\infty} f_G(g)\, dg \\
&= \int_{\frac{1}{2}}^{\infty} e^{-g}\, dg \\
&= -e^{-g}\Big|_{\frac{1}{2}}^{\infty} \\
&= e^{-\frac{1}{2}} \\
&= 0.61
\end{aligned}
$$

3. Explain the basics of wireless channel modelling (10 marks)

Or explain a) Delay spread

b) Coherence bandwidth

c) Doppler spread

d) coherence Time

<u>The Wireless Channel</u>

<u>Basics of wireless channel modelling</u>

The channel is characterized by multiple Non-Line-Of-Sight (NLOS) signal paths caused by scattering from objects in the environment (e.g., buildings, trees).

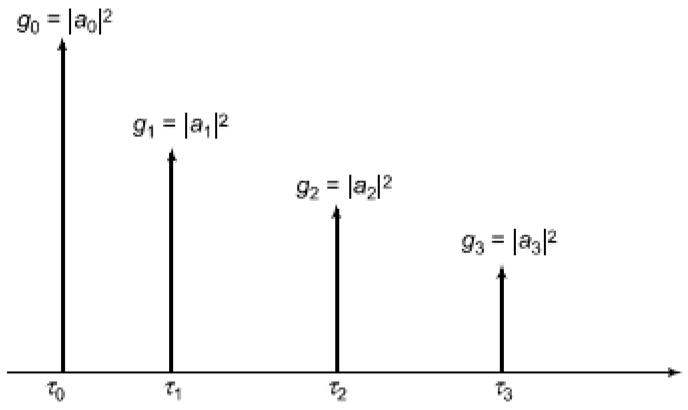


Figure 4.1 *Schematic of an $L = 4$ tap wireless channel profile*

Channel Model: The wireless channel's impulse response is modelled as a sum of delayed and attenuated signal paths:

$$h(t) = \sum_{i=0}^{L-1} a_i \delta(t - \tau_i)$$

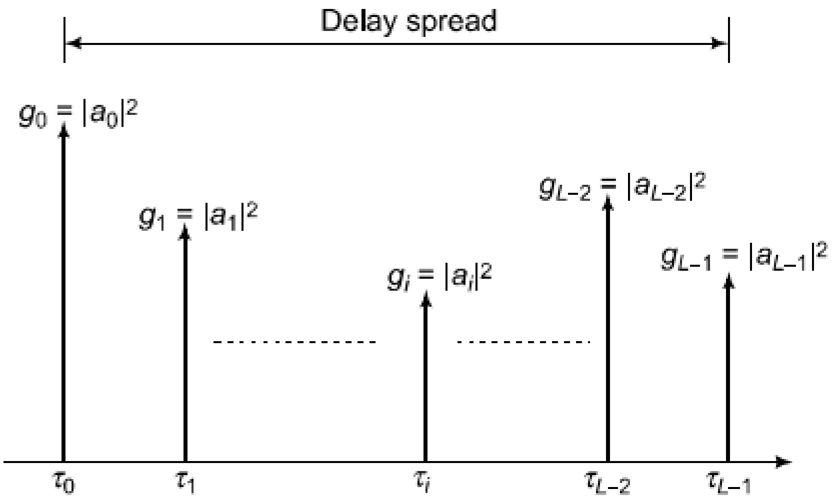
where ai is the complex attenuation and τi is the delay for the ith path.

Power **Profile:** The multipath power profile, ϕ(t), describes the distribution of received power over time. It is defined as the sum of the power gains $||a_i||^2$ (denoted as gi) at their respective delays:

$$\phi(t) = \sum_{i=0}^{L-1} g_i \delta(t - \tau_i)$$

1. **<u>Delay Spread:</u> Because the total signal power arrives incrementally over a range of delays (from τ_0 to τ_L−1), it is spread out in time. This phenomenon is called the delay spread.**

   In a wireless channel, power arrives and spread out over time due to multiple paths



2 *Schematic of a typical wireless channel power profile and delay spread*

Table 4.1 *Path gains and associated delays for an $L = 4$ multipath channel*

Gain	Delay
$\|a_0\|^2$	τ_0
$\|a_1\|^2$	τ_1
$\|a_2\|^2$	τ_2
$\|a_3\|^2$	τ_3

a) <u>Maximum delay spread</u>

The maximum delay spread is defined as the time difference between the arrival of the first and last significant multipath components:

$$\sigma_\tau^{\max} = \tau_{L-1} - \tau_0$$

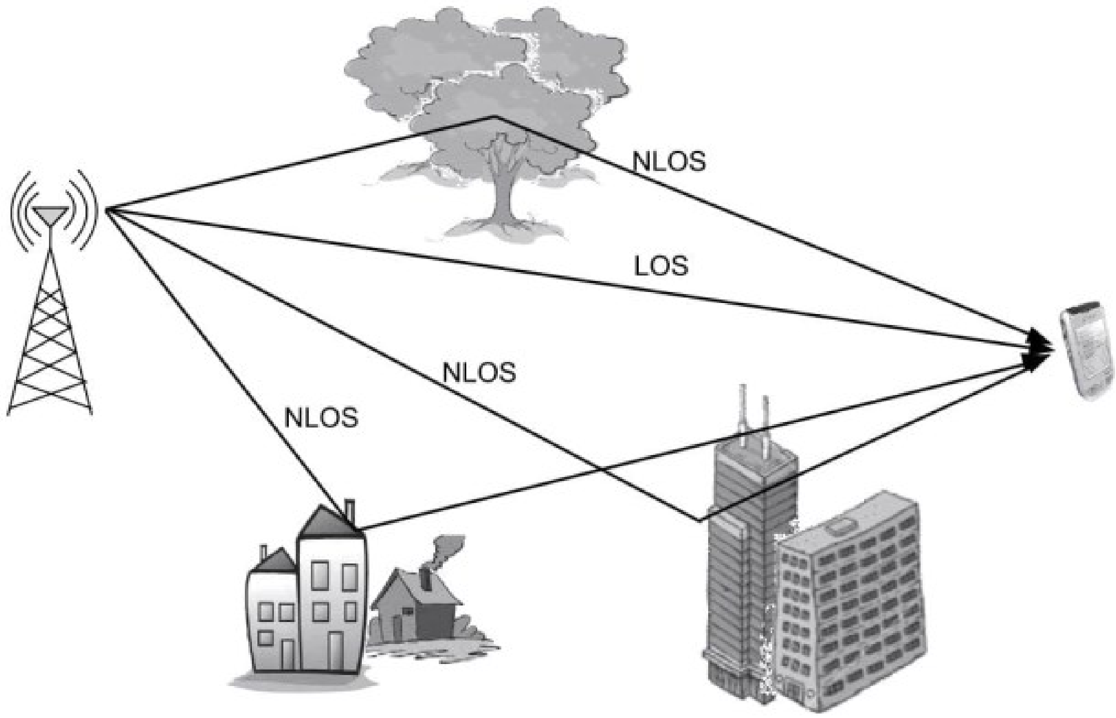


Figure 4.3 *Schematic of the wireless-propagation environment*

It measures the total time interval over which the signal's energy is received.

1. A larger $\sigma_\tau^{\max}$ indicates a "richer" scattering environment with greater differences in path lengths.
2. The delay spread depends only on the *difference* in path delays, not on the **absolute distance to the transmitter**. A receiver far away with only one path (like in free space) will have a delay spread of zero.
3. **Delay spread is fundamentally caused by the presence of multiple propagation paths.** Its value is determined by the richness of the scattering environment.

EXAMPLE 4.1

Consider an $L = 4$ multipath channel with the delays τ_0, τ_{L-1} corresponding to the first and last arriving paths, given as $\tau_0 = 0\ \mu s$ and $\tau_{L-1} = 5\ \mu s$. Such a wireless-channel power profile is shown schematically in Figure 4.4. What is the maximum delay spread σ_τ^{max} corresponding to this wireless channel?

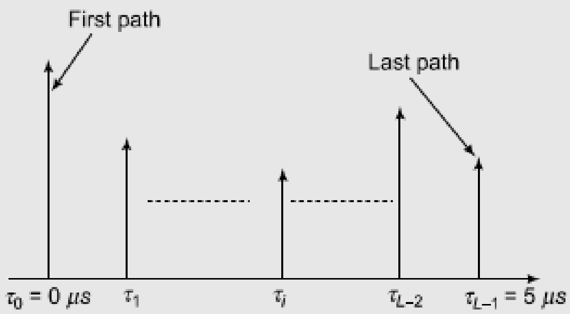


Figure 4.4 *Power profile for Example 4.1.*

Solution: By a simple application of the result in Eq. (4.2) for L multipath components, it can be seen that the maximum delay spread is

$$\sigma_\tau^{\max} = 5\ \mu s - 0\ \mu s = 5\ \mu s$$

b) <u>RMS delay spread</u>

The RMS delay spread automatically suppresses the contribution of weak, spurious multipath components, providing a better measure of the actual power spread of the signal.

- **Limitation of Maximum Delay Spread:** The maximum delay spread ($\sigma_\tau^{\max}$) can be misleading because it includes very weak, late-arriving paths that contribute little power. It does not weight the delays by their corresponding power.
- **Introduction of RMS Delay Spread:** The RMS delay spread is a more realistic indicator because it weights each path's delay by the fraction of total power it carries. It is calculated as the standard deviation of the delay values around their power-weighted mean.
- **Calculation Steps:**
  1. **Normalize Power Gains:** Calculate the fraction of total power in each path: $b_i = \frac{g_i}{\sum_{j=0}^{L-1} g_j}$.
  2. **Calculate Mean Delay:** Compute the power-weighted average delay: $\overline{\tau} = \sum_{i=0}^{L-1} b_i \tau_i$.
  3. **Calculate RMS Delay Spread:** Compute the standard deviation of the delays around this mean:

$$\sigma_\tau^{\text{RMS}} = \sqrt{\sum_{i=0}^{L-1} b_i (\tau_i - \overline{\tau})^2}$$

EXAMPLE 4.2

Consider the multipath power profile of a wireless channel shown in Figure 4.6 comprising $L = 4$ multipath components. Compute the RMS delay spread $\sigma_\tau^{\mathrm{RMS}}$ for this wireless channel.

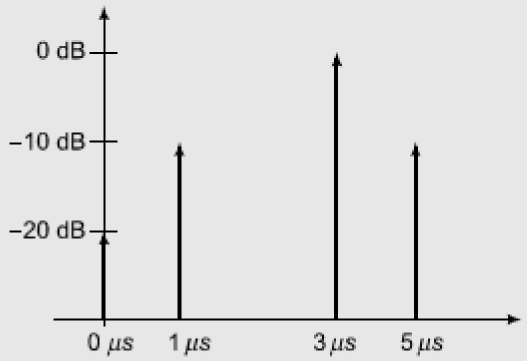


Figure 4.6 *Power profile for Example 4.2*

Solution: Consider the first path corresponding to $\tau_0 = 0\,\mu s$. The power associated with this path is $g\,(\,\mathrm{dB}) = -20$ dB. Hence, the linear power can be obtained as

$$10\log_{10} g_0 = -20 \text{ dB}$$
$$\Rightarrow \log_{10} g_0 = -2$$
$$\Rightarrow g_0 = 10^{-2} = 0.01$$

Also, the amplitude a_0 associated with this path can be derived as

$$a_0 = \sqrt{g_0} = 0.1$$

Thus, one can compute the corresponding power g_i and amplitude a_i for each of the $L = 4$ multipath components corresponding to $0 \le i \le 3$. These are listed in Table 4.2.

One can now compute the mean delay $\overline{\tau}$ for this channel as

$$\begin{aligned}\overline{\tau} &= \frac{\sum_{i=1}^{L-1} g_i\tau_i}{\sum_{i=0}^{L-1} g_i}\\ &= \frac{0.01\times 0 + 0.1\times 1 + 1\times 3 + 0.1\times 5}{0.01+0.1+1+0.1}\,\mu s\\ &= 2.9752\,\mu s\end{aligned}$$

Thus, the mean delay is $\overline{\tau} = 2.9752\,\mu s$. Employing the expression in Eq. (4.3), the RMS delay spread can be computed as

$$\begin{aligned}\sigma_\tau^{\mathrm{RMS}} &= \frac{0.01\times(0-2.9752)^2 + 0.1\times(1-2.9752)^2 + 1\times(3-2.9752)^2 + 0.1\times(5-2.9752)^2}{0.01+0.1+1+0.1}\\ &= 0.8573\,\mu s\end{aligned}$$

c). Average Delay Spread in Outdoor Cellular Channels

For a typical outdoor cellular scenario with a 1 km path difference between a direct path (2 km) and a scattered path (3 km), the maximum delay spread is calculated as:

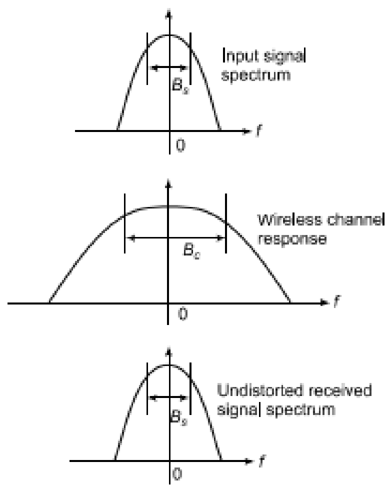
Outdoor: Delay spreads are on the order of **1–3 microseconds (μs)**.

Indoor: Due to much shorter distances (around 10 m), delay spreads are significantly smaller, on the order of **10–50 nanoseconds (nsns)**.

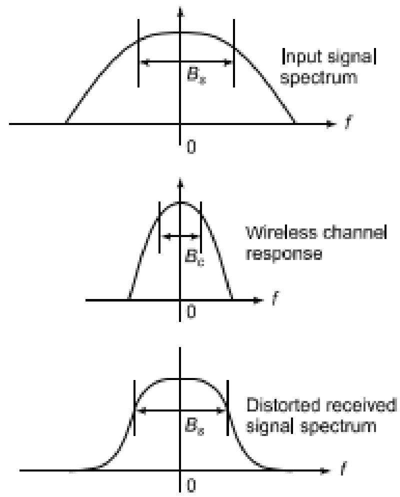
These approximate values are crucial for the practical design of wireless communication systems (like 3G/4G), as the delay spread is a key parameter that influences system performance and design choices. The example illustrates how the scale of the environment directly determines the scale of the delay spread.

2. Coherence bandwidth in wireless communication

Coherence Bandwidth (Bc) is a statistical measure of the range of frequencies over which a wireless channel can be considered "flat," meaning all frequency components experience approximately the same magnitude of fading and linear phase shift.



.11 *Signal bandwidth B_s less than coherence bandwidth B_c implying no distortion*



4.12 *Signal bandwidth B_s greater than coherence bandwidth B_c leading to distortion in spectrum of received signal*

- **Impact on Signal Transmission:**
    - **Flat Fading:** If the transmitted signal's bandwidth B_s is **less than or equal to** the coherence bandwidth ($B_s \leq B_c$), all frequency components of the signal experience the same attenuation. The signal is received without distortion.
    - **Frequency-Selective Fading:** If the signal's bandwidth is **greater than** the coherence bandwidth ($B_s > B_c$), different frequency components of the signal experience different attenuations. This causes distortion in the received signal.

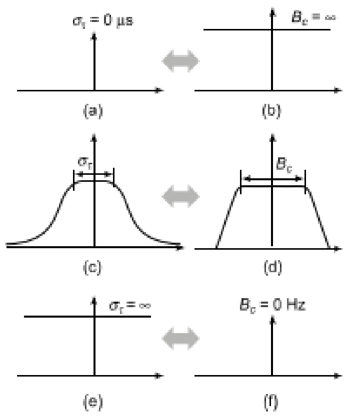


Figure 4.13 *Coherence bandwidth B_c variation with delay spread σ_τ*

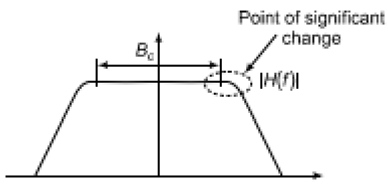


Figure 4.14 *Coherence bandwidth showing point of change of response*

- **Relationship with Delay Spread (σ_τ):** There is an inverse relationship between delay spread and coherence bandwidth. A larger delay spread (more time dispersion) results in a smaller coherence bandwidth (more rapid variations in the frequency response).
    - $\sigma_\tau = 0$ **(Single Path):** The channel has a flat frequency response ($H(f) = 1$) with infinite coherenc bandwidth. This is an ideal **flat-fading** channel.
    - σ_τ **increases:** The frequency response $H(f)$ begins to vary, and the coherence bandwidth B_c decreases.
    - $\sigma_\tau \to \infty$: The coherence bandwidth shrinks to zero.

- **Empirical Relationship:** An approximate formula for coherence bandwidth is derived based on the maximum delay τ_{L-1} of the channel:

$$B_c \approx \frac{1}{2\tau_{L-1}}$$

This formula indicates that the coherence bandwidth is inversely proportional to the maximum excess delay of the multipath components.

In essence, the coherence bandwidth is a frequency-domain measure of the channel's selectivity, directly determined by its time-domain dispersion (delay spread). It is a critical parameter for determining whether a channel will cause flat or frequency-selective fading for a given transmitted signal.

4. Relation between ISI and coherence bandwidth

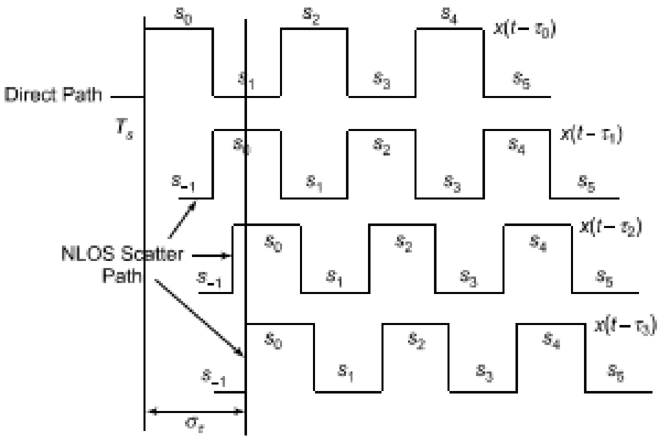


Figure 4.16 *Severe ISI caused by multiple scatter components*

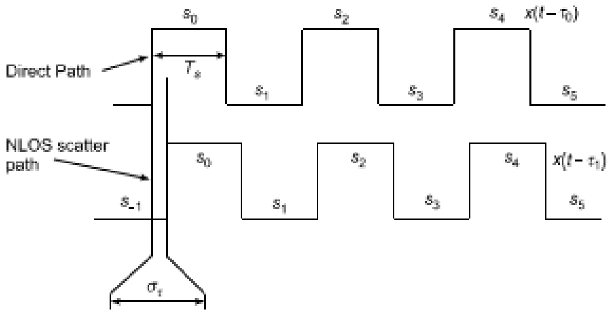


Figure 4.17 *Negligible ISI when $\sigma_\tau << T_s$*

1. **Cause of ISI:** ISI occurs when multiple copies of a signal, delayed in time (e.g., from a line-of-sight path and a scattered path), arrive at the receiver. If the **delay spread (T_d)** is comparable to or larger than the **symbol time (T_s)**, symbols overlap, causing interference.
2. Coherence Bandwidth (B_c) is a statistical measure of the **range of frequencies over which a wireless channel can be considered "flat"** or, in other words, over which the channel passes all spectral components with approximately equal gain and linear phase.
3. **ISI Criterion:** The condition for significant ISI is empirically given as $T_d \geq ½\, T_s$.
4. **Duality of Perspectives:**

- **Time Domain:** ISI is a **time-domain** phenomenon caused by a large delay spread relative to the symbol time.
- **Frequency Domain:** This is equivalent to the **signal bandwidth (B_s)** being larger than the channel's **coherence bandwidth (B_c)**, which is the condition for **frequency-selective fading**.

5. **Frequency-selective distortion and ISI are two interpretations of the same underlying channel behaviour.**

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5. . **Explain doppler impact and Doppler fading on wireless channel. (10 marks)**

Doppler fading in wireless systems

Doppler fading is a phenomenon in wireless communications where the frequency of a transmitted radio wave is perceived to change due to relative motion between the transmitter and receiver. This is a fundamental characteristic of mobile wireless systems that distinguishes them from static, wired communications.

1. **Cause:** Relative motion between the transmitter and receiver.
2. **Effect:** A shift in the perceived frequency of the signal.

   Higher perceived frequency if the transmitter and receiver are moving **towards** each other.

   Lower perceived frequency if they are moving **apart**.

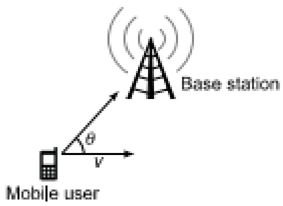

Figure 4.19 *Doppler scenario*

Doppler Shift Computation

The precise Doppler shift (f_d) is calculated using the following formula:

$$f_d = \left(\frac{v}{c}\cos\theta\right) f_c$$

Where:

- f_d = Doppler shift (the change in frequency)
- v = velocity of the mobile station
- c = speed of light (3×10^8 m/s)
- f_c = carrier frequency (the original transmitted frequency)
- θ = angle between the direction of motion and the line connecting the transmitter and receiver

<u>Doppler impact on wireless channel</u>

When the user moves, the signal paths keep changing, which shifts the frequency (Doppler effect). This makes the wireless channel coefficients vary with time, leading to fluctuations (fading) in the received signal.

User Movement and Signal Paths

In wireless communication, the transmitted signal doesn't travel along just one path.

It reflects off buildings, trees, vehicles, and other obstacles, creating multiple paths (multipath propagation).

Each path has a different distance → which introduces a different time delay before it reaches the receiver.

Effect of Movement: Frequency Shift (Doppler Effect)

The movement of the user (towards or away from the base station) causes a Doppler shift.

If moving towards the transmitter: the frequency appears slightly higher.

If moving away: the frequency appears slightly lower.

This shift depends on:

$$f_d = \frac{v\cos\theta}{c} f_c$$

where f_d = Doppler frequency, v = velocity, θ = angle of motion, f_c = carrier frequency.

Impact on Wireless Channel Coefficients

The wireless channel is represented by a coefficient h

which tells how the signal is modified when passing through the channel.

With Doppler, these coefficients are no longer fixed but change with time:

$$h(t) = \sum_{i=0}^{L-1} a_i e^{-j2\pi f_c \tau_i} e^{j2\pi f_d t}$$

That extra term $e^{j2\pi f_d t}$ shows that the channel phase is rotating with time.

EXAMPLE 4.5

Consider a vehicle moving at 60 miles per hour at an angle of $\theta = 30°$ with the line joining the base station. Compute the Doppler shift of the received signal at a carrier frequency of $f_c = 1850$ MHz.

Solution: To begin with, we convert the velocity v from units of miles per hour to the standard metres per second. Noting that a mile is equal to 1.61 km, the required velocity in meters per second can be derived as

$$\begin{aligned} 60\,\text{mph} &= 60 \times 1.61\,\text{kmph} \\ &= 60 \times 1.61 \times \frac{5}{18}\,\text{m/s} \\ &= 26.8\,\text{m/s} \end{aligned}$$

Employing the expression in Eq. (4.11), the Doppler shift f_d can be computed as

$$\begin{aligned} f_d &= \frac{26.8}{3 \times 10^8} \times \cos(30°) \times 1850 \times 10^6 \\ &= 143\ \text{Hz} \end{aligned}$$

Thus, the Doppler shift is $f_d = 143$ Hz. Further, since the mobile user is moving towards the base station, the Doppler shift is positive, i.e., the perceived frequency f_r is higher compared to the carrier frequency f_c and is given as $f_r = f_c + f_d = 1850\,\text{MHz} + 143\,\text{kHz}$.

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1. Write note on coherence time of wireless channels (10 marks)

Coherence Time (Tc) is time during which the channel (the connection between MS and BTS) stays roughly the same and predictable.

1. **Cause:** It is caused by the **time-varying** nature of the channel due to the **Doppler shift** resulting from user mobility.

2. **Definition:** The coherence time is the approximate time interval for which the channel can be assumed to be constant.
3. **Relationship to Doppler:** The coherence time (T_c) is inversely proportional to the **Doppler frequency (f_d)** and the **Doppler spread (B_d)**.

◦ A common empirical definition is:

$$T_c = \frac{1}{4 f_d^{\max}}$$

where $f_d^{\max} = \frac{v}{c} f_c$ is the maximum Doppler shift for a given velocity.

◦ It can also be expressed as:

$$T_c = \frac{1}{2B_d}$$

where $B_d = 2f_d$ is the Doppler spread.

Fast Changing Channel (High f_d): **A high Doppler frequency (from high user velocity) leads to a** short coherence time**. The channel changes rapidly.**

Slow Changing Channel (Low f_d): **A low Doppler frequency leads to a** long coherence time**. The channel remains relatively static.**

EXAMPLE 4.6

Consider the scenario described above in Example 4.5, i.e., a mobile user in a vehicle moving at 60 miles per hour. Compute the coherence time T_c at the carrier frequency $f_c = 1.85$ GHz.

Solution: To compute the coherence time T_c, we start by computing the maximum Doppler shift $f_d^{\max}$ corresponding to $\theta = 0°$. Following a procedure similar to the one in Example 4.5, this can be obtained as

$$\begin{aligned} f_d^{\max} &= \frac{26.8}{3 \times 10^8} \times 1850 \times 10^6 \\ &= 165 \text{ Hz} \end{aligned}$$

Hence, the corresponding Doppler spread is given as $B_d = 2 \times f_d^{\max} = 330$ Hz. Hence, the coherence time T_c is given from the relation in Eq. (4.13) as

$$\begin{aligned} T_c &= \frac{1}{2 \times 330} \\ &= 1.5 \text{ ms} \end{aligned}$$

Thus, as can be seen from the example above, the value of T_c in practical wireless systems, at vehicular velocities around 60 mph and carrier frequencies in the 2 GHz range is of the order of milliseconds (ms). Thus, the Doppler spread of a wireless system gives the wireless-system designer an idea of the rate of change of the wireless-channel coefficient. Also, it can be readily seen that a larger Doppler spread B_d corresponds to a smaller coherence time T_c leading to a faster rate of channel variation.